

بسم الله الرحمن الرحيم

$$\frac{\rho_o}{\rho_f} = 0.7$$

Q3

5 kg object displaces 3 kg of water. The acceleration is

$$a = \left(\frac{5 - 3}{5} \right) \times 10 = 9 \text{ down}$$

$$F_b = (m + 5)g$$

$$\rho_o V_o g = (m + 5)g$$

$$\rho_o \frac{m_o}{\rho_o} = m + 5$$

$$\rho_o \frac{m_o}{\rho_o} - m = 5$$

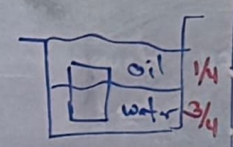
Q4

A ratio of ^① fluid to ^② object densities is 1.5

$$\frac{15 \rho_f}{10 \rho_o} = 1.5$$

$$a = \left(\frac{15 - 10}{10} \right) \times 10 \text{ up}$$

Q5

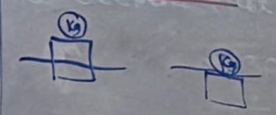


object of density ρ_{object} is suspended between two liquids of $\rho_{\text{water}} = 1000$ and $\rho_{\text{oil}} = 800$

$$\begin{aligned} \rho_{\text{object}} &= \frac{3}{4} \rho_{\text{water}} + \frac{1}{4} \rho_{\text{oil}} \\ &= \frac{3}{4} \times 1000 + \frac{1}{4} \times 800 = 950 \end{aligned}$$

Q6

Object of $\rho_o = 600$ if 5 kg is added + be full immersed in water



$$m_o = \frac{5}{\frac{\rho_f}{\rho_o} - 1}$$

$$m_o = \frac{5}{\frac{1000}{600} - 1} = 7.5 \text{ kg}$$

Q7

A ratio of object to fluid densities is 1.2

$$\frac{12 \rho_o}{10 \rho_f} = 1.2$$

$$a = \left(\frac{12 - 10}{12} \right) \times 10 \text{ down}$$

قسط المرفوع

$$\frac{\rho_o}{\rho_f} = 0.7$$

Q3

5 kg object displaces 3 kg of water. The acceleration is

$$a = \left(\frac{5 - 3}{5} \right) \times 10 = 4 \text{ down}$$

$$\frac{\rho_o}{\rho_f} = \frac{200}{1000} = \frac{2}{10}$$

Q4

A ratio of ⁽¹⁾ fluid to ⁽²⁾ object densities is 1.5

$$\frac{\rho_f}{\rho_o} = 1.5$$

$$a = \left(\frac{15 - 10}{10} \right) \times 10 \text{ up}$$

Q5



object of density ρ_{object} is suspended between two liquids of $\rho_{\text{water}} = 1000$ and $\rho_{\text{oil}} = 800$

$$\rho_{\text{object}} = \frac{3}{4} \rho_{\text{water}} + \frac{1}{4} \rho_{\text{oil}}$$

$$= \frac{3}{4} \times 1000 + \frac{1}{4} \times 800 = 800$$

Q6

Object of $\rho_o = 600$ if 5 kg is added to be fully immersed in water



$$m_o = \frac{5}{\frac{\rho_o}{\rho_f} - 1}$$

$$m_o = \frac{5}{\frac{600}{1000} - 1} = 7.5 \text{ kg}$$

Q7

A ratio of object to fluid densities is 1.2

$$\frac{\rho_o}{\rho_f} = 1.2$$

$$a = \left(\frac{12 - 10}{12} \right) \times 10 \text{ down}$$

نفس المسألة (2)

⊖
-2.5 μC

$r = 1m$

⊕
6 μC

$q = -5nC$ at $(0.3m, -0.2m)$
 $E = ?$ at $P (-0.1, -0.6m)$

$$r_x = 0.3 - (-0.1) = 0.4$$

$$r_y = -0.2 - (-0.6) = 0.4$$

⊕	⊖
$r_x = x_p - x_q$	$r_x = x_q - x_p$
$r_y = y_p - y_q$	$r_y = y_q - y_p$

$$r = \sqrt{r_x^2 + r_y^2}$$

$$\theta = \tan^{-1}(r_y/r_x)$$

$$E = k \frac{q}{r^2}$$

$$r = 1m$$

⊖

$$-2.5 \mu C$$

⊕

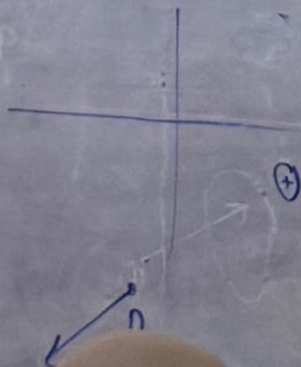
$$6 \mu C$$

$$q = 5nC \text{ at } (0.3m, -0.2m)$$

$$E = ? \text{ at } P (-0.1, -0.6m)$$

$$r_x = -0.1 - 0.3 = -0.4$$

$$r_y = -0.6 - (-0.2) = -0.4$$



⊕

$$r_x = x_p - x_q$$

$$r_y = y_p - y_q$$

⊖

$$r_x = x_p - x_q$$

$$r_y = y_p - y_q$$

$$r = \sqrt{r_x^2 + r_y^2}$$

$$\theta = \tan^{-1}(r_y/r_x)$$

$$E = k \frac{q}{r^2}$$

مسألة

$r = 1m$

$\dot{P} \quad x \quad (-)$
 $-2.5 \mu C$

$(+)$
 $6 \mu C$

$$x = \frac{1}{\sqrt{\frac{6}{2.5} - 1}} = 1.82$$

$\leftarrow r = 3m \rightarrow$

$(-)$
 $-32e$

$\dot{P} \quad x \quad (-)$
 $-2e$

$$x = \frac{3}{\sqrt{\frac{32}{2} + 1}} = 0.6$$